DOC-21 | CONTROLLED SOURCE SYNCHRONIZATION

Cross-reference-only revision notice

v1.0-RC2 | 11 September 2026 | SOURCE-OF-TRUTH SYNCHRONIZATION ONLY

Control	Selected value
Revision	v1.0-RC1 -> **v1.0-RC2**
Semantic change	**NONE.** This revision changes source version/count/path references only.
Wave-2 semantic authorities	DOC-07 RC2 | DOC-08 RC2 | DOC-15 RC8 | DOC-16 RC5
Canonical HTTP companion	contracts/http/BLOG-DOC-15_openapi_v1.0-RC8.yaml
OpenAPI SHA-256	bf577a463b4c5d8fdceddc2087f53216426ef88d67d8e1a0d22ccf819751cd00
RC8 estate	113 operations | 86 paths | 221 schemas/helpers | 1,311 resolved internal refs | 0 unresolved refs 12 AUTH | 2 PUBLIC/PREVIEW | 63 SITE | 36 GLOBAL

DOCUMENT-SPECIFIC SYNCHRONIZATION

• QA source references now consume DOC-15 v1.0-RC8 and DOC-16 v1.0-RC5. The QA21 ID estate remains exactly 180 IDs; no QA ID is added or retired.

• QA21-API-001 and QA21-REG-005 count assertions are updated to: 113 operations | 86 paths | 221 schemas/helpers | 1,311 resolved internal refs | 0 unresolved refs; 12 AUTH | 2 PUBLIC/PREVIEW | 63 SITE | 36 GLOBAL.

• QA21-API-002 applies to all 113 canonical operations. QA21-API-009 now verifies API15-AUTH-001..012 against RC8. QA21-REG-009 now expects 12 AUTH operations aligned with DOC-19 mechanics.

• All remaining QA procedures, evidence rules, severity and stop-the-line semantics remain unchanged.

SUPERSESSION AND HISTORICAL CLASSIFICATION

The inherited body that follows is retained unchanged in substantive page content. Any inherited non-owning reference to DOC-15 RC7/RC6, the old 112/85/217/1,297 estate, 11 AUTH operations, DOC-16 RC4, or the RC7 YAML filename is classified **HISTORICAL / SUPERSEDED REFERENCE** and is not normative in this revision. All other inherited owned semantics remain in force.

Occurrence ledger: RC7/old HTTP: inherited printed pages 1, 3, 4, 9, 10, 20, 21, 22, 23, 24; RC4: inherited printed pages 1, 3, 4, 21, 22, 24.

IMPLEMENTATION IMPACT: NONE. The accepted NAKSHATRA_MASTER_WAVE2_HANDOFF application/database/API behavior is unchanged by this document synchronization.

BLOG-DOC-21

QA, Multi-Site Isolation, Concurrency & Acceptance Test Specification

Release-blocking QA authority for deterministic end-to-end verification of the centralized 14-site blogging platform.

14	36	112	1
public sites	Admin Page IDs	API operations	release gate

RELEASE PRINCIPLE

A candidate release passes only when every applicable QA21 test is PASS, every required evidence artifact is attached, no tenant/security/data-integrity leak occurs, and all upstream numeric or semantic acceptance gates remain satisfied. There is no waiver path inside this specification for an unresolved release-blocking failure.

Document control	Value
Status	REVIEW CANDIDATE - built against user-designated finalized/frozen BLOG-DOC-01 through BLOG-DOC-20; freeze after DOC-21 approval
Version	1.0-RC1
Date	10 September 2026
Upstream authority set	BLOG-DOC-01 through BLOG-DOC-20; current late-chain references: DOC-15 RC7, DOC-16 RC4, DOC-17 RC2, DOC-18 RC2, DOC-19 RC2, DOC-20 RC1
Primary authority	Release-blocking test architecture; deterministic QA21 IDs; test environments/data; execution/evidence rules; isolation, authorization, concurrency, workflow, security, SEO, media, worker, accessibility, performance and recovery acceptance
Explicit non-authority	Does not create features, permissions, capabilities, pages, DB objects, API contracts, security mechanisms, algorithms, visual design, providers, repositories or deployment topology; tests only verify frozen behavior.

DEVELOPER LOCK TARGET

A competent QA or implementation team must not guess which tenant pairs to test, what constitutes a release-blocking failure, how concurrency races are reproduced, which evidence is required, whether wrong-site disclosure is acceptable, how upstream acceptance IDs map to executable tests, or which DOC-20 thresholds define pass/fail. Those decisions are locked here.

Table of Contents / Document Map

Section	Subject
1	Purpose, authority and conflict rule
2	Frozen upstream baseline and acceptance inheritance
3	Release-gate doctrine and severity
4	Deterministic QA21 identifier system
5	Test architecture and environment classes
6	Deterministic test data and 14-site collision fixtures
7	Execution, repeatability and evidence rules
8	Result states, flake handling and defect linkage
9	Multi-site isolation suite
10	Database/RLS and same-site integrity suite
11	HTTP/OpenAPI contract suite
12	Authorization and permission-attack suite
13	Admin flow, site switching and multi-tab suite
14	Public page, routing and PageData suite
15	Concurrency, simultaneous editing and autosave suite
16	Publishing, scheduling, Preview and lifecycle suite
17	Media/storage/CDN suite
18	SEO, search, sitemap and redirect suite
19	Redis/cache/worker/job suite
20	Security/auth/session/MFA/audit/privacy suite
21	Responsive and accessibility suite
22	Performance, load and noisy-neighbor suite
23	Backup, PITR, DR and recovery suite
24	Regression and frozen-count guard suite
25	Release execution pipeline and stop-the-line rules
26	Cross-document consistency audit
27	Traceability and coverage ledger
28	Freeze-readiness checklist
Appendix A	Canonical QA21 suite registry
Appendix B	Upstream release-acceptance coverage ledger
Appendix C	Evidence artifact manifest
Appendix D	One-page release execution reference

ONE SOURCE OF TRUTH

DOC-21 owns how frozen requirements are tested and what evidence is release-blocking. It never changes what the requirement means. If a test cannot be written without redefining an upstream requirement, the owning upstream document must be reopened rather than “clarified” inside a test.

1. Purpose, Authority and Conflict Rule

BLOG-DOC-21 converts the frozen product, isolation, persistence, routing, flow, data, frontend, API, SEO, media, worker, security and performance contracts into one deterministic release-verification system. It is the final QA authority for the behavior already defined in DOC-01 through DOC-20.

CONFLICT RULE

A QA test may choose fixtures, timing controls, fault injection and observation techniques, but it may not create a new business rule. When upstream authorities conflict, testing stops and the conflict is reported; expected behavior is never invented from convenience.

1.1 Authority partition

Decision	Authority
Product / 14-site estate / scope	DOC-01
Users, roles, 69 permissions and authorization	DOC-02
161 capabilities	DOC-03
Trusted SiteContext / isolation	DOC-04
Public/Admin page inventories	DOC-05 / DOC-06
Post/body/publishing lifecycle	DOC-07
PostgreSQL schema/RLS/seeds	DOC-08
Domain/routes/slug/redirect routing	DOC-09
Admin/public execution states	DOC-10 / DOC-11
Public components/accessibility	DOC-12
Public PageData	DOC-13
14 frontend application boundaries	DOC-14
HTTP/OpenAPI transport	DOC-15 RC7
SEO/search/sitemap/redirect computation	DOC-16 RC4
Media/storage/image/CDN	DOC-17 RC2
Jobs/queues/Redis/cache	DOC-18 RC2
Security/auth/session/audit/privacy	DOC-19 RC2
Performance/scale/backup/DR/observability	DOC-20 RC1
Executable release QA	DOC-21 - this document

2. Frozen Upstream Baseline and Acceptance Inheritance

Authority	Frozen acceptance inheritance into DOC-21
DOC-01..03	14 sites/one platform model, exact scope, 19 modules, 161 capabilities and no feature invention are regression guards.
DOC-04	All 75 ISO-REQ requirements become executable cross-tenant negative tests; isolation failure is release-blocking.
DOC-05/06	Exactly 14 public Page IDs and 36 Admin Page IDs; page-estate tests may not create generic Pages or hidden Admin surfaces.
DOC-07	52 ACC-CNT assertions plus five Post states, 12 blocks, scheduling, autosave, revisions and optimistic concurrency become workflow tests.
DOC-08	88 DB-ACC assertions, 33 tables, 20 FORCE-RLS site-owned tables, 69 permission seeds and same-site FK/RLS rules become DB integration tests.
DOC-09	73 routing acceptance assertions become canonical host/path/slug/redirect/404/410 tests.

Authority	Frozen acceptance inheritance into DOC-21
DOC-10/11	76 AC10 and 56 AC11 assertions plus page-specific QA seeds become page-flow/state/multi-tab/public-runtime tests.
DOC-12/13	74 QA12 + 72 AC13 assertions become accessibility/component/PageData contract tests.
DOC-14	36 QA14 assertions become independent-frontend/import-boundary integration tests.
DOC-15 RC7	70 AC15 assertions; 112 operations, 85 paths, 217 schemas and 1,297 resolved refs become contract tests.
DOC-16 RC4	104 normative QA16 assertions become SEO/search/sitemap/redirect tests without algorithm drift.
DOC-17 RC2	83 QA17 assertions become upload/storage/variant/CDN/privacy/delete/replace tests.
DOC-18 RC2	69 QA18 assertions become queue/job/dedupe/Redis/retry/reconciliation tests.
DOC-19 RC2	67 QA19 assertions become auth/MFA/session/CSRF/CORS/audit/privacy/secret/recovery tests.
DOC-20 RC1	72 QA20 assertions and LT20-01..12 scenarios become exact load/SLO/RPO/RTO/DR gates.

INHERITANCE RULE

An upstream release-blocking assertion remains release-blocking even if DOC-21 covers it through a parameterized test family rather than a one-to-one test. Appendix B records the complete coverage mapping. A PASS in DOC-21 cannot weaken an upstream threshold.

3. Release-Gate Doctrine and Severity

Class	Meaning	Release action
RB-BLOCKER	Tenant leak, auth bypass, wrong-site disclosure, data corruption/lost update, incorrect publish/Preview exposure, unrecoverable backup/restore failure, or failed frozen acceptance criterion.	STOP release; no waiver in DOC-21.
RB-FAIL	Any deterministic QA21 expected result not met.	STOP release until fixed and rerun.
BLOCKED	Environment/evidence unavailable, test cannot execute.	Treat as not passed; release cannot proceed.
FLAKY	Pass/fail varies without code/config change.	Not a pass. Quarantine only after reproducer + owner; release-critical test still must achieve deterministic PASS.
PASS	Expected result satisfied with required evidence.	Eligible for gate only after suite completeness checks.

- No percentage-based pass threshold exists for functional/isolation/security suites: required tests are 100% PASS.
- Performance tests use DOC-20 numeric gates exactly; “close enough” is FAIL.
- A defect fixed after a failed release test requires rerun of the failed test, its dependency-neighbor tests and the defined regression smoke set.
- Manual exploratory evidence may supplement but cannot replace a deterministic automated release test when automation is technically possible.

4. Deterministic QA21 Identifier System

```
QA21-{FAMILY}-{NNN}
Example: QA21-ISO-004
```

FAMILY is stable; NNN is zero-padded and never reused after publication.

Family	Scope
ISO	Multi-site isolation / wrong-site / collision
DB	PostgreSQL, RLS, same-site integrity
API	OpenAPI/HTTP contract
PERM	Authorization/permission attacks
ADM	Admin flow/site switching/multi-tab
PUB	Public routing/PageData/runtime
CONC	Simultaneous edits/autosave/version races
LIFE	Publish/schedule/Preview/lifecycle
MED	Media/storage/CDN
SEO	SEO/search/sitemap/redirect
WRK	Workers/Redis/cache/queues
SEC	Security/auth/MFA/session/audit/privacy
A11Y	Responsive/accessibility
PERF	Load/SLO/noisy-neighbor
DR	Backup/PITR/restore/recovery
REG	Frozen counts/schema/import/regression guards

PARAMETERIZATION RULE

A single QA21 ID may mandate an exhaustive parameter matrix. The QA ID identifies the invariant; each parameter instance produces a child execution key such as QA21-ISO-003[SITE-01->SITE-02,Post]. Every required child instance must pass.

5. Test Architecture and Environment Classes

Environment ID	Purpose	Required characteristics
ENV21-UNIT	Pure deterministic logic tests	No network; canonical validators/algorithms; cannot substitute for RLS/isolation integration.
ENV21-INT	Service + real dependency integration	Real PostgreSQL with FORCE RLS, Redis, durable broker, search adapter, S3-compatible object store; production-equivalent auth boundaries.
ENV21-E2E	Browser/API release candidate	All 14 frontend builds + Central Admin + API; TLS/Host routing; no mocked tenant authority.
ENV21-LOAD	Performance/fairness/chaos	Production-shaped topology; telemetry enabled; data scale from DOC-20; isolated from development noise.
ENV21-DR	Backup/restore drills	Disposable isolated restoration target; no accidental writes to production; real backup artifacts and recovery controls.

5.1 Dependency realism

- RLS tests must execute against PostgreSQL roles/policies equivalent to production; an in-memory database is invalid evidence.
- Queue dedupe/retry tests require a broker/Redis implementation that supports the semantics selected later by engineering; mocks may unit-test code but do not close integration acceptance.
- Browser tests must send real Host values and cannot inject trusted SiteContext directly.
- Security tests exercise the actual session/CSRF/CORS/step-up implementation selected for the candidate, not a bypass test mode.

- Performance numbers are accepted only from ENV21-LOAD with observability and fault-injection timestamps captured.

6. Deterministic Test Data and 14-Site Collision Fixtures

Fixture	Locked QA construction
TENANT-SET	All canonical SITE-01..SITE-14 exist. Isolation-negative suites execute directed foreign-site pairs: 14 x 13 = 182 source->foreign combinations where applicable.
SLUG-COLLISION	Each site owns an eligible Post slug `shared-article`, Category slug `shared-category`, Tag slug `shared-tag`, Author slug `shared-author`; content values include unique site markers.
MEDIA-COLLISION	Each site uploads a different image using the same original filename `shared-image.jpg`; metadata includes unique site marker.
NAV/REDIRECT-COLLISION	Each site has the same menu_key and same redirect source path pointing to different same-site targets.
POST-STATES	Per site: DRAFT, SCHEDULED, PUBLISHED, ARCHIVED, TRASH plus URL-history/permanent-delete fixture where allowed.
PERMISSION-USERS	QA principals are created from existing permission keys only: no-role, view-only, exact-action, missing-prerequisite, SITE-assigned, other-site-assigned, GLOBAL-only, designated Super Admin.
CONCURRENCY	Two independent browser sessions and two API clients target the same Post version; a third tab targets another site with same slug.
SEARCH/SITEMAP	Each site contains overlapping terms but unique marker text; sitemap rows share route shapes while hosts differ.
PERF-DATA	At least 100,000 Posts total and the DOC-20 hot-tenant distribution/load profile.

FIXTURE SAFETY

Test fixtures may create content and roles allowed by frozen capabilities, but they may not alter the permission registry, create a new lifecycle state, disable RLS, add a hidden test endpoint, inject site_id as authority, or loosen security to make tests easier.

7. Execution, Repeatability and Evidence Rules

1. Materialize deterministic fixture manifest including site keys, resource IDs, account IDs, current versions, expected slugs and timestamps.
2. Record candidate build/revision, OpenAPI SHA-256, DB migration/seed version, frontend build identifiers and environment identifier.
3. Execute prerequisite health checks without bypassing production readiness semantics.
4. Run suites in the gate order defined in Section 25. Stop immediately on tenant/security/data-corruption blockers.
5. Capture evidence at source: HTTP transcripts, DB assertions under correct role, browser screenshots/DOM/accessibility output, queue/Redis/search observations, metrics/traces, backup restore reports.
6. Store result record keyed by QA21 ID + parameter instance + candidate revision.
7. For failures, preserve correlation/request IDs and timestamps sufficient to join API, audit, worker and trace evidence.
8. Rerun only after defect/config change is identified; a blind retry does not convert a failure into a pass.

7.1 Required result record

```
qa_id: QA21-ISO-003
instance: SITE-01->SITE-02/Post
candidate: <immutable build id>
environment: ENV21-E2E
started_at / finished_at: UTC
status: PASS | FAIL | BLOCKED | FLAKY
evidence: [http, browser, db, audit, worker, metrics as applicable]
source_trace: [DOC-04 ISO-REQ-..., DOC-15 AC15-...]
```

defect_id: required when FAIL/FLAKY

8. Result States, Flake Handling and Defect Linkage

- PASS requires all child matrix instances pass; partial matrix completion is BLOCKED, not PASS.
- Any cross-site leak, private/unpublished leak, auth bypass, secret exposure or backup restore that violates ownership is immediately RB-BLOCKER.
- Flaky release-critical tests remain blocking until root cause is fixed or the implementation is made deterministic; increasing retry count to hide a race is prohibited.
- A defect may close only with evidence from the original failing instance plus the required regression neighborhood.
- Expected 401/403/404/409/410/422/429 outcomes are test successes when they match frozen contracts; tests do not treat all non-2xx responses as failures.

9. Multi-Site Isolation Suite

These tests are always release-blocking and run before functional suites. Directed tenant matrices are exhaustive wherever a foreign-site object can be supplied.

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
QA21-ISO-001	All 14 sites; same Post slug locator/name exists where legal.	For every directed site pair, enter SITE-A context then request/mutate SITE-B Post slug; repeat through browser/API/derived surface relevant to public same slug.	No SITE-B payload/metadata/count/timing hint is disclosed; no mutation occurs; same SITE-A object remains correct. FAIL on any cross-site data or authority. Trace: DOC-04 ISO-REQ-001..075; DOC-05 PUB-ISO; DOC-09 ACC
QA21-ISO-002	All 14 sites; same Category ID/slug locator/name exists where legal.	For every directed site pair, enter SITE-A context then request/mutate SITE-B Category ID/slug; repeat through browser/API/derived surface relevant to category.	No SITE-B payload/metadata/count/timing hint is disclosed; no mutation occurs; same SITE-A object remains correct. FAIL on any cross-site data or authority. Trace: DOC-04; DOC-08 same-site FKs/RLS; DOC-10 category QA
QA21-ISO-003	All 14 sites; same Tag ID/slug locator/name exists where legal.	For every directed site pair, enter SITE-A context then request/mutate SITE-B Tag ID/slug; repeat through browser/API/derived surface relevant to tag.	No SITE-B payload/metadata/count/timing hint is disclosed; no mutation occurs; same SITE-A object remains correct. FAIL on any cross-site data or authority. Trace: DOC-04; DOC-08; DOC-10 tag QA
QA21-ISO-004	All 14 sites; same Author ID/slug locator/name exists where legal.	For every directed site pair, enter SITE-A context then request/mutate SITE-B Author ID/slug; repeat through browser/API/derived surface relevant to author.	No SITE-B payload/metadata/count/timing hint is disclosed; no mutation occurs; same SITE-A object remains correct. FAIL on any cross-site data or authority. Trace: DOC-04; DOC-08; DOC-10 author QA
QA21-ISO-005	All 14 sites; same Media ID/filename locator/name exists where legal.	For every directed site pair, enter SITE-A context then request/mutate SITE-B Media ID/filename; repeat through browser/API/derived surface relevant to media.	No SITE-B payload/metadata/count/timing hint is disclosed; no mutation occurs; same SITE-A object remains correct. FAIL on any cross-site data or authority. Trace: DOC-04; DOC-17 QA17-001..083
QA21-ISO-006	All 14 sites; same navigation/menu key locator/name exists where legal.	For every directed site pair, enter SITE-A context then request/mutate SITE-B navigation/menu key; repeat through browser/API/derived surface relevant to navigation.	No SITE-B payload/metadata/count/timing hint is disclosed; no mutation occurs; same SITE-A object remains correct. FAIL on any cross-site data or authority. Trace: DOC-04; DOC-10 navigation QA; DOC-13 navigation
QA21-ISO-007	All 14 sites; same redirect ID/source locator/name exists where legal.	For every directed site pair, enter SITE-A context then request/mutate SITE-B redirect ID/source; repeat through browser/API/derived surface relevant to	No SITE-B payload/metadata/count/timing hint is disclosed; no mutation occurs; same SITE-A object remains correct. FAIL on

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
		redirect.	any cross-site data or authority. Trace: DOC-04; DOC-09; DOC-16 QA16-RED-01..16
QA21-ISO-008	All 14 sites; same search term locator/name exists where legal.	For every directed site pair, enter SITE-A context then request/mutate SITE-B search term; repeat through browser/API/derived surface relevant to search.	No SITE-B payload/metadata/count/timing hint is disclosed; no mutation occurs; same SITE-A object remains correct. FAIL on any cross-site data or authority. Trace: DOC-04; DOC-16 QA16-SRCH-01..20
QA21-ISO-009	All 14 sites; same sitemap resource locator/name exists where legal.	For every directed site pair, enter SITE-A context then request/mutate SITE-B sitemap resource; repeat through browser/API/derived surface relevant to sitemap.	No SITE-B payload/metadata/count/timing hint is disclosed; no mutation occurs; same SITE-A object remains correct. FAIL on any cross-site data or authority. Trace: DOC-04; DOC-16 QA16-SM-01..14
QA21-ISO-010	All 14 sites; same cacheable route locator/name exists where legal.	For every directed site pair, enter SITE-A context then request/mutate SITE-B cacheable route; repeat through browser/API/derived surface relevant to cache.	No SITE-B payload/metadata/count/timing hint is disclosed; no mutation occurs; same SITE-A object remains correct. FAIL on any cross-site data or authority. Trace: DOC-04 cache isolation; DOC-18 Redis namespace
QA21-ISO-011	All 14 sites; same job entity identity locator/name exists where legal.	For every directed site pair, enter SITE-A context then request/mutate SITE-B job entity identity; repeat through browser/API/derived surface relevant to worker.	No SITE-B payload/metadata/count/timing hint is disclosed; no mutation occurs; same SITE-A object remains correct. FAIL on any cross-site data or authority. Trace: DOC-04 worker context; DOC-18 QA18-001..069
QA21-ISO-012	All 14 sites; same preview post/grant locator/name exists where legal.	For every directed site pair, enter SITE-A context then request/mutate SITE-B preview post/grant; repeat through browser/API/derived surface relevant to preview.	No SITE-B payload/metadata/count/timing hint is disclosed; no mutation occurs; same SITE-A object remains correct. FAIL on any cross-site data or authority. Trace: DOC-04 Preview isolation; DOC-19 Preview security
QA21-ISO-013	Unknown Host plus known site slugs.	Request known slugs using unmapped/foreign Host and attempt host-header alias tricks.	Neutral non-disclosing failure; no guessed tenant shell/data. FAIL if any site is inferred. Trace: DOC-04 host resolution; DOC-09; DOC-11; AC15
QA21-ISO-014	Disable/archive a site with primed caches/search/sitemap.	Prime all derived surfaces, change site lifecycle through authorized flow, then continue public and worker requests.	Ordinary public resolution fails per upstream lifecycle; stale cache/search/sitemap cannot keep site public. Trace: DOC-04 lifecycle; DOC-05 PUB-ISO-09; DOC-18/20
QA21-ISO-015	Pooled DB connections exercised across alternating sites.	Alternate hundreds of SITE-01/SITE-02 requests through the same pool; intentionally reuse connections.	No transaction inherits prior app.site_id; every query is scoped to current SET LOCAL context. Trace: DOC-04; DOC-08 DB-ACC; QA20-005
QA21-ISO-016	Global aggregate user with limited site assignments.	Open allowed GLOBAL network view, inspect rows, drill into site-owned targets including unassigned site.	Aggregate shows only authorized rows; drilldown constructs target CTX-ADMIN-SITE and denies unassigned site. Trace: DOC-02; DOC-04; DOC-10 Global All-Posts; AC15-046

10. PostgreSQL, RLS and Same-Site Integrity Suite

Database tests prove the defense-in-depth layer independently of UI and service filtering.

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
QA21-DB-001	Production-equivalent DB roles/policies.	Enumerate all 20 site-owned canonical tables; verify FORCE RLS, policy presence and non-BYPASSRLS runtime role.	All 20 enforce RLS; ordinary runtime cannot bypass. FAIL on any unprotected site-owned table. Trace: DOC-08 DB-ACC-001..088; DOC-04
QA21-DB-002	SITE-01 and SITE-02 rows exist in every applicable site-owned table.	Execute SELECT/UPDATE/DELETE under SET LOCAL SITE-01 and probe SITE-02 IDs.	Only SITE-01 rows visible/mutable; foreign rows behave unavailable. Trace: DOC-08 RLS acceptance; DOC-04
QA21-DB-003	Same-site relationship fixtures plus foreign IDs.	Attempt Post->Category/Tag/Author/Media/Related and navigation/home relationships using foreign-site IDs.	Constraint/service validation rejects; no mixed-site relationship commits. Trace: DOC-08 same-site composite FK direction; DOC-07
QA21-DB-004	Pooled transaction begins without app.site_id.	Run ordinary site-owned repository call with missing scope.	Fails closed; missing scope never returns all rows. Trace: DOC-04; DOC-08
QA21-DB-005	Valid Post current version N.	Run two CAS updates expected_version=N concurrently.	Exactly one commits; other gets deterministic conflict; final version/state is coherent. Trace: DOC-07 ACC-CNT; DOC-08; DOC-15 AC15-019
QA21-DB-006	Permission seed baseline.	Count permission rows and scope split; attempt wildcard/deny/mutation.	Exactly 69 = 44 SITE + 25 GLOBAL; registry is immutable through product API. Trace: DOC-02; DOC-08; DOC-10 QA10-G-PERM-06; AC15-016
QA21-DB-007	Canonical schema migrated.	Inventory tables and ownership class.	Exactly 33 canonical tables = 20 site-owned + 13 global/system; no QA implementation adds hidden product table. Trace: DOC-08; DOC-01
QA21-DB-008	Audit-producing mutations.	Perform allowed/denied mutations across SITE/GLOBAL.	Successful state changes produce immutable audit rows with correct context; denied request never produces a successful-mutation audit event. Trace: DOC-08 audit_events; DOC-10; DOC-19
QA21-DB-009	Deleted/retired taxonomy/author/media targets.	Attempt relationship save/publish using non-eligible related records.	Validation/state rules reject or omit exactly as upstream; no dangling public projection. Trace: DOC-07; DOC-08; DOC-13
QA21-DB-010	Restore/PITR fixture database.	After restore drill, rerun RLS/seeds/state/version checks.	RLS/FORCE RLS, 69 seeds, ownership, Post states/versions and audit integrity remain valid. Trace: DOC-20 QA20-049..050

11. HTTP / OpenAPI Contract Suite

This suite treats openapi.yaml as executable transport contract and the DOC-15 PDF as human authority from the same source.

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
QA21-API-001	DOC-15 RC7 PDF and delivered OpenAPI YAML.	Parse YAML; enumerate operations/paths/schemas/$refs and operation IDs.	Exactly 112 operations = 2 PUBLIC/PREVIEW + 11 AUTH + 63 SITE + 36 GLOBAL; 85 paths; 217 schemas; 1,297 refs; zero unresolved refs. Trace: DOC-15 RC7 AC15-001,043,049
QA21-API-002	All 112 operations.	For each operation compare method/path/success status/auth context/scope/permissions/rate/idempotency/concurrency metadata with PDF registry.	Zero drift. Any mismatch blocks release. Trace: DOC-15 RC7 AC15-001..070
QA21-API-003	Schema corpus with valid/invalid generated samples.	Contract-test required/optional fields, enums, lengths, unknown top-level	Valid samples accepted; invalid samples rejected with frozen status/error

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
		properties and oneOfs.	semantics. Trace: DOC-15 AC15-042..050
QA21-API-004	Public PageData roots.	Request each of 14 Page IDs through canonical routes; validate direct DOC-13 body, omission rules, [] rules and no transport wrapper.	Schemas match DOC-13 exactly; no site_id/internal fields/null drift. Trace: DOC-13 AC-001..072; DOC-15 AC15-008..009
QA21-API-005	Versioned Post mutation endpoints.	Submit stale and current expected_version; retry with same/different Idempotency-Key fingerprint.	409 on stale/different-fingerprint; identical replay does not duplicate side effects. Trace: DOC-15 AC15-019,033..035
QA21-API-006	All Admin list/search endpoints.	Exercise page/page_size bounds and empty pages.	Admin default 1/50 max100; invalid bounds 400; no tenant-blind list. Trace: DOC-15 AC15-036
QA21-API-007	Public collection/search.	Exercise pagination bounds and blank search.	Public default 1/20 max50; blank query returns NO_QUERY + [] with no pagination. Trace: DOC-13; DOC-15 AC15-010,036
QA21-API-008	Rate-limitable operations.	Drive each rate class to ceiling.	429 includes Retry-After + RateLimit headers; rate class matches registry; no auth/scope bypass. Trace: DOC-15 AC15-037; DOC-19
QA21-API-009	AUTH family API15-AUTH-001..011.	Validate exact methods/paths/statuses/schemas/logical auth contexts against RC7.	All 11 canonical auth operations are present and aligned; mechanics remain DOC-19. Trace: DOC-15 RC7; DOC-19 QA19-054
QA21-API-010	Media endpoints after RC6 carry forward.	Verify search/filter/sort/pagination, variants, usage_count, title/source_url and replace metadata authorization.	All media transport reconciliation behavior remains intact in RC7. Trace: DOC-15 AC15-055..059; DOC-17 RC2

12. Authorization and Permission-Attack Suite

Every security-sensitive action is tested by direct invocation, not only by UI visibility.

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
QA21-PERM-001	For every protected operation, principal lacks exactly the required action permission.	Invoke API directly even if UI hides control.	Denied server-side; no mutation or protected payload. Trace: DOC-02; DOC-10 action auth; DOC-15 AC15-015
QA21-PERM-002	SITE assignment present but action permission absent.	Open SITE page/API and invoke mutation.	Assignment alone never grants capability. Trace: DOC-02; DOC-04
QA21-PERM-003	GLOBAL permission only, no SITE assignment/permission.	Attempt site content read/mutation.	GLOBAL role does not imply SITE content authority. Trace: DOC-02; AC15-017
QA21-PERM-004	SITE permission on SITE-01 only.	Attempt same operation on SITE-02 by path/ID substitution.	Denied/non-disclosed; no cross-site privilege carryover. Trace: DOC-02; DOC-04
QA21-PERM-005	Role bundle omits a required prerequisite .view key.	Attempt to create/assign invalid role combination and execute action.	Invalid bundle rejected; no silent prerequisite completion. Trace: DOC-02 prerequisite rules; DOC-08 permission_prerequisites
QA21-PERM-006	Category/Author editor has base edit but lacks seo.view or seo.edit.	Submit ordinary+SEO fields in one request.	Entire mutation rejected atomically; ordinary fields not partially saved. Trace: DOC-15 AC15-051..053; DOC-16 API QA
QA21-PERM-007	Media replacer lacks media.edit.	Replace binary while also supplying metadata.	Entire replace rejected atomically; binary generation not switched. Trace: DOC-15 AC15-059; DOC-17
QA21-PERM-008	Tag editor lacks tags.delete.	Invoke merge.	Merge denied; composite tags.view+edit+delete is required.

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
			Trace: DOC-02; DOC-10 Tag Merge; AC15-027
QA21-PERM-009	Revision viewer/restorer missing one required component permission.	Invoke revision restore directly.	Denied with zero durable change. Trace: DOC-02; DOC-07; DOC-10; DOC-15
QA21-PERM-010	Bulk action includes at least one unauthorized target/action.	Submit multi-target bulk command.	Security prevalidation prevents unauthorized partial commit. Trace: DOC-10 bulk; DOC-15 AC15-025
QA21-PERM-011	Non-Super Admin attempts privilege escalation via role/site assignment.	Assign role outside actor privilege ceiling or self-escalate.	Rejected; actor cannot grant authority they do not possess. Trace: DOC-02 privilege ceiling; DOC-15 user/role APIs
QA21-PERM-012	Permission Registry page/API.	Try create/edit/delete key or submit wildcard.	Exactly 69 fixed keys read-only; no mutation surface. Trace: DOC-02; DOC-10 QA10-G-PERM-06; AC15-016

13. Admin Flow, Site Switching and Multi-Tab Suite

The Admin browser is tested as a concurrent client with stale requests, dirty state and independent tabs.

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
QA21-ADM-001	Dirty SITE editor in SITE-01.	Switch Active Website to SITE-02; choose cancel, then discard, then save paths in separate runs.	Explicit dirty handling occurs; no unsaved SITE-01 values migrate into SITE-02. Trace: DOC-10 common site-switch/dirty rules
QA21-ADM-002	Two browser tabs on different sites.	Change Active Website in tab A; continue read/mutation in tab B.	Each tab retains own request context; no browser-global tenant variable controls authority. Trace: DOC-10 multi-tab; DOC-04
QA21-ADM-003	Slow SITE-01 request in flight.	Switch page to SITE-02 before response completes.	Stale SITE-01 completion is discarded and never rendered into SITE-02. Trace: DOC-10 ST10-STALE-CONTEXT
QA21-ADM-004	GLOBAL page open with Active Website selector set.	Change selector and perform GLOBAL action.	GLOBAL authority/target does not retarget from selector; site drilldown establishes new CTX-ADMIN-SITE. Trace: DOC-10 GLOBAL common rules
QA21-ADM-005	All 36 Admin Page IDs.	Parameterize page entry with allowed, denied, empty, error and wrong-target conditions.	Each page follows its QA10 entry/scope/action/state/navigation family; no extra operational page appears. Trace: DOC-06 AC-001..050; DOC-10 AC10-001..076 + page QA
QA21-ADM-006	Recoverable API failure on SITE page.	Force one dependency 503 then restore; use page retry.	Retry stays in same page/site/target; no context broadening. Trace: DOC-10 ST10-ERROR-RETRY; DOC-15
QA21-ADM-007	Read-only Admin list/filter page.	Navigate away/back and refresh.	Filter state may reset per page contract but no domain mutation occurs. Trace: DOC-10 page contracts
QA21-ADM-008	Destructive action awaiting confirmation.	Change target/site/session authority before confirming.	Commit reauthorizes current target; stale confirmation cannot mutate prior/foreign target. Trace: DOC-10 ST10-CONFIRMING; DOC-19 step-up
QA21-ADM-009	Dashboard/list contains site-local links.	Follow links under normal and stale contexts.	Destination loads only under correct SiteContext and permission; no UI-derived authority. Trace: DOC-10 handoffs; DOC-04
QA21-ADM-010	Admin session revoked while page remains open.	Attempt next protected read/mutation.	Next request detects revocation and denies; cached UI state is not authority.

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
			Trace: DOC-19 session freshness; DOC-15

14. Public Page, Routing and PageData Suite

Public tests execute across all 14 applications and preserve both route identity and data-projection authority.

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
QA21-PUB-001	All 14 sites x 14 public Page IDs where route applies.	Load canonical routes, refresh, back/forward and deep links.	Correct Page ID/Host/SiteContext and state; no site-to-site frontend/data import. Trace: DOC-05; DOC-11 AC11-001..056; DOC-14
QA21-PUB-002	Published/Draft/Scheduled/Archived/ Trash fixtures.	Request ordinary Article route for each state.	Only PUBLISHED public-eligible renders; non-public states do not leak. Trace: DOC-05; DOC-07; DOC-11
QA21-PUB-003	Existing Category/Tag/Author with zero eligible Articles.	Load archive route.	Valid EMPTY page, not 404. Trace: DOC-11 page-specific QA; DOC-13
QA21-PUB-004	Retired/missing/wrong-site taxonomy/author.	Load route before/after valid redirect setup.	Unavailable/404 unless valid redirect wins; no hidden metadata. Trace: DOC-09; DOC-11
QA21-PUB-005	Known wrong path and unknown host.	Request 404 cases.	Safe NotFoundPageData; unknown host has no guessed tenant shell. Trace: DOC-09; DOC-13; AC15-012
QA21-PUB-006	Former public Post locator after permanent delete with/without redirect.	Request historical URL.	Redirect wins when valid; otherwise narrow 410 where DOC-09 says; no deleted metadata. Trace: DOC-09; DOC-13; AC15-013
QA21-PUB-007	Optional media/related content missing.	Break optional dependency without changing primary Article eligibility.	Valid primary content remains; site-local missing-media/omission behavior, never foreign fallback. Trace: DOC-11; DOC-12; DOC-13; DOC-17
QA21-PUB-008	Public search blank/results/empty fixtures.	Submit whitespace, matching and nonmatching queries; paginate.	NO_QUERY/RESULTS/EMPTY semantics and authoritative server order preserved. Trace: DOC-11; DOC-13; DOC-16
QA21-PUB-009	Browser request started on SITE-A then navigate to SITE-B domain before completion.	Delay API response and complete after navigation.	Old async completion cannot overwrite current site/route/query state. Trace: DOC-11 common state rules; DOC-14
QA21-PUB-010	Each frontend build.	Static/import analysis plus runtime smoke.	No site-to-site application import; shared approved contracts/client/utilities only; page shell/header/footer/assets remain site-local. Trace: DOC-14 QA14-001..036

15. Concurrency, Simultaneous Editing and Autosave Suite

Concurrency tests force races with barriers/fault injection; sequential simulations are insufficient evidence for race-specific cases.

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
QA21-CONC-001	Two editors load Post version N.	Editor A saves; Editor B saves stale version N.	A succeeds and increments; B receives 409 VERSION_CONFLICT with zero overwrite. Trace: DOC-07; DOC-10 QA10-S-POST-3-09; AC15-019

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
QA21-CONC-002	Editor autosave base_version N while durable save advances to N+1.	Submit/receive autosave after durable save.	Autosave remains recovery state and cannot overwrite durable server state or create revision/publish. Trace: DOC-07 autosave; DOC-15 AC15-020
QA21-CONC-003	Two autosave requests with same Idempotency-Key.	Retry identical request after ambiguous client timeout.	Original outcome replays; no duplicate semantic effect. Trace: DOC-15 idempotency
QA21-CONC-004	Revision restore and ordinary edit race on same Post.	Release both concurrently with same expected_version.	Exactly one commits; loser gets conflict; no hybrid body/state. Trace: DOC-07 revisions/concurrency; DOC-15
QA21-CONC-005	Publish and edit race.	Submit eligible publish and stale edit concurrently.	Version/state fences produce one coherent order; no lost update or duplicate revision. Trace: DOC-07 WF; DOC-18
QA21-CONC-006	Archive/Trash and public read race.	Hold public request across mutation commit.	After authoritative mutation, new public requests follow eligibility; stale cache cannot restore public state beyond frozen convergence. Trace: DOC-07; DOC-18; DOC-20
QA21-CONC-007	Two users edit Category/Tag/Author/Media.	Concurrent mutations target same resource using current API semantics.	No cross-site/partial unauthorized state; service/database constraints remain valid. Trace: DOC-08; DOC-10; DOC-15
QA21-CONC-008	Bulk Post action with mixed versions.	Submit authorized targets including stale versions.	Security prevalidation first; business/version outcomes follow API contract without unauthorized partial execution. Trace: DOC-10 bulk; DOC-15
QA21-CONC-009	Idempotent command times out client-side after server commit.	Replay same key then different body with same key.	Same key/same fingerprint replays; different fingerprint 409; no double command. Trace: DOC-15 AC15-033..035
QA21-CONC-010	SITE switch while save in flight.	Start save in SITE-A then switch UI to SITE-B.	Server result remains SITE-A-bound and UI discards stale completion; SITE-B untouched. Trace: DOC-10; DOC-04
QA21-CONC-011	Search reindex plus incremental Post changes.	Start full reindex; mutate/publish Posts during build.	Shadow generation activates atomically and queued incremental work catches up; no missed current change. Trace: DOC-16; DOC-18
QA21-CONC-012	Sitemap build while multiple changes occur.	Advance desired_seq repeatedly during build.	One build publishes atomically; follow-up occurs if desired_seq > built_seq. Trace: DOC-18 sitemap coordination

16. Publishing, Scheduling, Preview and Lifecycle Suite

Lifecycle tests validate state, side effects and worker convergence together; UI-only state checks are insufficient.

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
QA21-LIFE-001	Valid DRAFT Post.	Publish through authorized command.	Transitions exactly per DOC-07; public/search/sitemap/cache/audit effects are triggered without duplicate business mutation. Trace: DOC-07 ACC-CNT; DOC-03 cross-feature; DOC-18
QA21-LIFE-002	Valid DRAFT plus local schedule/timezone input.	Schedule and inspect persisted UTC result + delayed worker.	SCHEDULED with server-derived scheduled_for_utc; no client-authoritative UTC override. Trace: DOC-07; DOC-15 SchedulePostInput; DOC-18

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
QA21-LIFE-003	Scheduled Post generation/version A.	Reschedule to generation/version B; allow old delayed job to fire.	Old job is STALE_NOOP; only current due identity can publish. Trace: DOC-18 schedule identity
QA21-LIFE-004	Scheduled Post before due time.	Run worker early.	No early publication; job waits/retries per schedule semantics. Trace: DOC-07; DOC-18
QA21-LIFE-005	Scheduled Post then unschedule.	Fire old delayed job afterward.	Old job no-ops; Post remains upstream-approved non-public state. Trace: DOC-07; DOC-18
QA21-LIFE-006	Scheduled Post then manual early publish.	Let old scheduled job arrive later.	No duplicate publish/revision; later job stale/no-op. Trace: DOC-07; DOC-18
QA21-LIFE-007	140 Posts due in same 10-second window across sites.	Execute LT20-05.	At least 99% publish within 60s of due; none early or duplicate. Trace: DOC-20 LT20-05 / QA20
QA21-LIFE-008	Draft working copy and Preview permission.	Create preview handoff, consume once on correct host, then replay/use wrong host/post/site.	Preview renders privately/no-store/noindex; grant not in PageData; replay/wrong binding denied. Trace: DOC-07; DOC-15; DOC-19
QA21-LIFE-009	TRASH Post.	Restore.	Returns to approved non-public state; never silently PUBLISHED. Trace: DOC-07; AC15-022
QA21-LIFE-010	Post eligible for permanent delete with historical URL.	Delete and request former URL.	Terminal delete semantics, audit, URL-history/redirect/410 behavior and derived cleanup follow upstream; no deleted metadata leak. Trace: DOC-07; DOC-09; DOC-16

17. Media, Object Storage and CDN Suite

Media tests combine API, object storage, derivative verification and public CDN behavior while never exposing private origin credentials in evidence.

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
QA21-MED-001	Valid/invalid raster corpus.	Upload JPEG/PNG/static WebP/static AVIF plus wrong signature, animated/multiframe, oversize, >12k dimension and >80MP samples.	Only DOC-17 allowlisted/valid samples accepted; invalids rejected before unsafe processing. Trace: DOC-17 QA17-001..083
QA21-MED-002	Valid image containing GPS/EXIF/comments.	Upload and inspect private original vs every anonymous public derivative.	Private original may preserve source metadata; public native_webp/variants are sanitized and expose no GPS/EXIF/comment metadata. Trace: DOC-17 privacy reconciliation
QA21-MED-003	Valid upload.	Wait for processing; enumerate variant records and files.	Only fixed 16-key catalog; no upscaling; dimensions/MIME/byte_size match verified objects. Trace: DOC-17
QA21-MED-004	Small source below larger derivative targets.	Process all variants.	No derivative upscales; unavailable larger variants omitted rather than fabricated. Trace: DOC-17
QA21-MED-005	Square-thumbnail source with off-center content.	Generate square thumbnail repeatedly.	Deterministic center crop (0.5,0.5); no authored focal-point field exists. Trace: DOC-17
QA21-MED-006	Media referenced by Article/navigation/home/SEO.	Attempt delete.	Deletion safety rechecks authoritative usages; in-use object cannot be unsafely purged. Trace: DOC-17; DOC-08 media_usages
QA21-MED-007	Media current generation G1 referenced publicly.	Replace with G2 and force G2 variant failure before native_webp verify.	Generation switch aborts; G1 remains current/public-safe. Trace: DOC-17 replacement safety;

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
			DOC-18
QA21-MED-008	Replacement succeeds through native_webp then delay other variants.	Read PublicMedia during regeneration.	media_id/references preserved; native_webp is safe baseline; later variants appear only after verified. Trace: DOC-17; DOC-13 PublicMedia
QA21-MED-009	Media list with collisions/usages.	Exercise DOC-15 search/mime/usage/page/page_size/sort.	Server-side same-site filtering; deterministic tie-break; usage_count available without cross-site/N+1 leakage. Trace: DOC-15 AC15-055..059
QA21-MED-010	Delete media then inspect CDN/object store/backups over lifecycle.	Verify immediate public revoke, 7-day primary purge eligibility with rechecks and backup residual policy.	No deleted media remains anonymously public; purge/backup behavior matches DOC-17/20. Trace: DOC-17; DOC-20 QA20-070

18. SEO, Search, XML Sitemap and Redirect Suite

Discovery tests verify computed behavior without redefining the algorithms frozen in DOC-16.

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
QA21-SEO-001	Article with all fallback fields combinations.	Parameterize seo_title/title and description fallbacks.	Exact DOC-16 hierarchy; no fabricated marketing copy. Trace: DOC-16 QA16-SEO-01..20
QA21-SEO-002	Category/Tag/Author empty vs non-empty public archive.	Resolve robots/indexability.	Category/Tag/Author index only when frozen policy/eligible-Article gates pass; page>1 noindex,follow. Trace: DOC-16 QA16-SEO
QA21-SEO-003	Search/Preview/404/410 pages.	Inspect robots/canonical/sitemap.	Always NOINDEX and absent from sitemap; no accidental canonical to private/system page. Trace: DOC-16; DOC-05
QA21-SEO-004	Canonical override corpus: valid local/alias/external HTTPS plus invalid credentials/fragments/private/system.	Save/read/render canonical projection.	Validation and normalization exactly follow DOC-16; external canonical never changes routing/ownership. Trace: DOC-16 QA16-SEO/DATA/API
QA21-SEO-005	Article OG field combinations and missing media.	Render OpenGraph.	og_title/description/image fallback order exact; image always same-site public-safe. Trace: DOC-16; DOC-17
QA21-SEO-006	All public page kinds.	Inspect structured_data arrays.	Only DOC-16 allowlist types; no raw Admin-authored JSON-LD/script injection. Trace: DOC-16
QA21-SEO-007	Breadcrumb-enabled page types.	Inspect trail destinations across same/foreign site fixtures.	Correct same-site PublicDestination only; no wrong-site breadcrumb. Trace: DOC-16; DOC-13
QA21-SEO-008	ACTIVE site with mixed eligible/noneligible URLs and external canonicals.	Fetch sitemap index/shards.	Only 200, public-eligible, index=true, local route-canonical page-1 URLs; max 25,000/shard; deterministic canonical URL order. Trace: DOC-16 QA16-SM-01..14
QA21-SEO-009	robots.txt on ACTIVE and site-level NOINDEX; alias host.	Fetch resources.	Preview disallowed, canonical sitemap declared; site-level NOINDEX disallows /; alias canonicalizes first. Trace: DOC-16
QA21-SEO-010	Search corpus with normalization collisions.	Query NFKC/case/punctuation/whitespace variants.	Normalization exact; no stemming/transliteration/synonym/wildcard/operator semantics. Trace: DOC-16 QA16-SRCH-01..20
QA21-SEO-011	Search results with equal/near scores.	Compute server result order and repeat across runs/reindex.	Exact weighted ranking and tie-break order remains deterministic; site scope

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
			applied before ranking. Trace: DOC-16
QA21-SEO-012	Redirect graph with live route, manual redirect, history, 410, cycles and >3 internal edges.	Resolve each source including self/cross-site/private destinations.	Precedence and 301/302/308 rules exact; cycles/self/unsafe/>3 rejected/suppressed; live route wins. Trace: DOC-09; DOC-16 QA16-RED-01..16

19. Redis, Cache, Worker and Background-Job Suite

Worker tests observe queue state, Redis state machine, canonical DB state and derived outputs together.

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
QA21-WRK-001	Durable queue/broker active.	Accept API15-S-094 reindex then terminate API process.	202 occurs only after durable acceptance; job survives ordinary process restart. Trace: DOC-15; DOC-18
QA21-WRK-002	Any site-owned job payload with forged foreign site/entity identity.	Deliver to worker.	Worker re-resolves lifecycle, CTX-WORKER-SITE, RLS and ownership; payload is never authorization. Trace: DOC-04; DOC-18
QA21-WRK-003	Redeliver successful job N times.	Force broker redelivery after side effect.	At-least-once converges without duplicate business mutation; ACK semantics correct. Trace: DOC-18 QA18
QA21-WRK-004	Search sync Post then mutate same Post within 24h.	Ensure second source fingerprint differs; enqueue while earlier TERMINAL dedupe remains.	New current-generation sync is not suppressed; index converges to latest content. Trace: DOC-18 RC2 QA18-065
QA21-WRK-005	Category/Tag/Author term changes with unchanged posts.version.	Run entity refresh fan-out.	Entity source fingerprint causes affected Posts to recompute search fingerprints and refresh. Trace: DOC-18 RC2 QA18-066
QA21-WRK-006	Create CLAIMED dedupe key then fail broker enqueue.	Test known failure and ambiguous crash paths.	Known failure compare-deletes claim; ambiguous claim expires <=60s and reconciliation can reclaim; CLAIMED is never durable proof. Trace: DOC-18 RC2 QA18-067..068
QA21-WRK-007	Broker accepted message but producer crashes before ENQUEUED CAS.	Allow worker/reconciliation to proceed.	Worker may repair state; duplicate enqueue is safe; no lost work or duplicate business effect. Trace: DOC-18 RC2
QA21-WRK-008	Lease holder stalls beyond renewal while replacement attempt starts.	Release stale attempt token afterward.	Stale attempt cannot release newer lease; DB/version/generation fence remains final correctness guard. Trace: DOC-18
QA21-WRK-009	Retryable dependency errors for each of 7 retry profiles.	Inject failures and record delays/attempt/timeouts.	Max attempts/base/cap/timeout and deterministic jitter match DOC-18 exactly; terminal failures DLQ. Trace: DOC-18 retry registry
QA21-WRK-010	Expected stale/no-op job and security-invariant mismatch.	Execute both.	Stale/no-op ACKs without DLQ; security invariant may dead-letter immediately without cross-site broadening. Trace: DOC-18
QA21-WRK-011	Redis flush/restart with canonical DB intact.	Run traffic/reconciliation.	Editorial truth unchanged; cache/search/sitemap coordination rebuilds same-site; no auth decision depends on Redis cache data. Trace: DOC-18; DOC-20 LT20-07
QA21-WRK-012	Hot SITE-A backlog plus runnable jobs for other sites.	Run 10k mixed backlog.	Priority/convergence targets met; SITE-A does not hold >50% shared queue concurrency >60s while others wait. Trace: DOC-18; DOC-20 LT20-06 / noisy-

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
			neighbor

20. Security, Authentication, Sessions, Audit, Privacy and Secrets Suite

Security cases use the real RC7 authentication transport while treating DOC-19 as the mechanics authority.

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
QA21-SEC-001	Active/disabled/invited accounts.	Attempt login with valid/invalid password and account states.	Generic/non-disclosing auth behavior; only eligible ACTIVE authentication establishes Admin session. Trace: DOC-02; DOC-19
QA21-SEC-002	Password setup/reset corpus.	Exercise invitation setup, authenticated change, reset request/completion, expired/replayed token.	Argon2id policy and one-time/expiry semantics exact; replay invalid; no account enumeration. Trace: DOC-19; API15-AUTH-003..006
QA21-SEC-003	Designated Super Admin.	Attempt privileged session without completed MFA.	MFA mandatory; no hidden bypass. Trace: DOC-19
QA21-SEC-004	TOTP enrollment/verification/recovery codes.	Use correct/incorrect/replayed recovery code and clock-window boundaries.	Only valid proof succeeds; recovery code one-time behavior preserved; secrets never returned afterward improperly. Trace: DOC-19; API15-AUTH-007..009
QA21-SEC-005	Authenticated session.	Idle past 30m and separately pass 12h absolute expiry.	Session rejected after frozen idle/absolute limits. Trace: DOC-19
QA21-SEC-006	Session valid then account disabled/role/site assignment changed.	Use next protected request.	Current account/assignment/permission is rechecked; stale session does not retain revoked authority. Trace: DOC-19; DOC-02
QA21-SEC-007	Two sessions for same user.	Terminate current/target session through canonical operation; test old bearer.	Revoked session cannot be used afterward; other behavior follows frozen session rules. Trace: DOC-19; API15-AUTH-010
QA21-SEC-008	Sensitive/destructive operations across L1/L2/L3 requirements.	Attempt without required step-up, with valid proof, expired proof and proof from another session.	Exact step-up gate enforced; proof bound appropriately and cannot be replayed across session/context. Trace: DOC-19; API15-AUTH-011
QA21-SEC-009	Cookie-authenticated mutation.	Send missing/invalid CSRF and wrong/missing Origin, then valid request.	CSRF + exact Origin policy blocks forged requests; valid request passes if otherwise authorized. Trace: DOC-19
QA21-SEC-010	Credentialed cross-origin requests.	Probe unapproved origins and wildcard CORS behavior.	No wildcard credentialed origin; only approved exact origin behavior. Trace: DOC-19
QA21-SEC-011	Public/Admin/Preview response matrix.	Inspect required security headers and cache headers.	Headers/TLS/cache behavior exact; Admin/Preview success no-store. Trace: DOC-19; DOC-15
QA21-SEC-012	Credential-abuse scenarios plus DOC-15 quotas.	Exceed auth/reset/token/session risk thresholds and endpoint ceilings.	Security throttles may be stricter but never raise DOC-15 ceilings; 429 behavior remains correct. Trace: DOC-15; DOC-19
QA21-SEC-013	Malicious text/HTML/URL/embed/upload metadata corpus.	Store/render through Admin/public paths.	Validated/encoded output; no raw script/style/unknown block or unsafe redirect/embed execution. Trace: DOC-07/12/13/15/19
QA21-SEC-014	Secret canary values in env/session/MFA/reset/Preview/object-store credentials.	Exercise errors, logs, traces, Redis, queue payloads, PageData and audit.	No secret/canonical private credential leaks into prohibited surfaces. Trace: DOC-17/18/19/20
QA21-SEC-015	State-changing operations.	Correlate API request, audit event and worker events where applicable.	Required audit event exists with privacy-minimized details; immutable; failed auth is not recorded as successful

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
			mutation. Trace: DOC-19 audit dictionary; DOC-10
QA21-SEC-016	Incident/recovery drill for session-store/secret compromise.	Revoke/rotate per DOC-19 and retry old artifacts.	Old sessions/secrets/grants cannot regain authority; recovery preserves tenant isolation and audit trail. Trace: DOC-19; DOC-20 QA20-054,057

21. Responsive and Accessibility Suite

The suite verifies frozen semantic/responsive accessibility requirements; it does not pre-empt later site-specific visual/mockup authority.

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
QA21-A11Y-001	All public page types on every frontend.	Run automated accessibility scan plus keyboard/manual semantic checks.	Implementation targets WCAG 2.2 AA baseline; no release-blocking upstream accessibility violation. Trace: DOC-12 A11Y-12-01; QA12
QA21-A11Y-002	All interactive controls.	Operate using keyboard only, including menus/search/pagination/modal/disclosures.	All actions operable; visible focus preserved. Trace: DOC-12 A11Y-12-02
QA21-A11Y-003	All PageData/render states.	Inspect H1 and heading hierarchy.	Exactly one page H1; semantic heading order follows structure, not visual sizing. Trace: DOC-12 A11Y-12-03
QA21-A11Y-004	Header/nav/main/footer/search/system surfaces.	Inspect accessibility tree/landmarks.	Appropriate landmarks or equivalent labels exist without duplicate ambiguity. Trace: DOC-12 A11Y-12-04
QA21-A11Y-005	Icon-only controls and links.	Inspect accessible names and destination/action meaning.	Programmatic names exist; link text/name identifies purpose. Trace: DOC-12 A11Y-12-05
QA21-A11Y-006	Loading/error/no-results/Preview/private states.	Trigger each state without relying on color.	State is programmatically communicated; not color/animation-only. Trace: DOC-12 A11Y-12-06
QA21-A11Y-007	Mobile nav/disclosure/modal.	Open/close by keyboard/Escape; track focus.	Focus enters predictably and restores to trigger; ordinary navigation does not steal focus. Trace: DOC-12 A11Y-12-07
QA21-A11Y-008	Media informative/redundant/missing cases.	Inspect alt/caption/credit and broken-media handling.	Alt semantics match DOC-12/17; filenames/broken paths are never announced. Trace: DOC-12 A11Y-12-08; DOC-17
QA21-A11Y-009	Search/pagination dynamic update.	Change query/page by keyboard and observe announcements/focus.	Results/state changes announced without unexpected focus movement. Trace: DOC-12 A11Y-12-09
QA21-A11Y-010	Representative 360x800, 768x1024, 1440x900 QA viewports; reduced-motion preference.	Exercise all 14 Page IDs and key Admin interactions; compare route/action/state semantics.	No viewport changes authority, order meaning, eligibility or actions; content remains usable; reduced-motion preference respected. Viewports are QA fixtures, not product breakpoints. Trace: DOC-11 responsive invariants; DOC-12 A11Y-12-10; DOC-14

22. Performance, Load and Noisy-Neighbor Suite

Performance tests use exactly the DOC-20 reference profile and thresholds; this document defines automation/evidence, not new SLOs.

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
QA21-PERF-001	ENV21-LOAD; 14 sites; >=100k Posts.	Run LT20-01: 300 public RPS for 30m.	All public latency/availability SLOs pass; zero cross-site leakage. Trace: DOC-20 LT20-01; QA20-006..013
QA21-PERF-002	Same.	Run LT20-02: 600 public RPS for 5m.	No correctness loss; horizontal scaling stays within DB connection budget. Trace: DOC-20 LT20-02
QA21-PERF-003	Hot-tenant 60/20/20 distribution.	Run LT20-03.	Other sites keep >=90% baseline throughput and <=1.5x normal p95 latency. Trace: DOC-20 LT20-03
QA21-PERF-004	Admin load mix.	Run LT20-04: 25 RPS + mutations with concurrency races.	No lost updates; expected_version conflicts deterministic; Admin latency SLOs pass. Trace: DOC-20 LT20-04; QA20-009
QA21-PERF-005	140 due publishes.	Run LT20-05.	>=99% within 60s after due; no early/duplicate publish. Trace: DOC-20 LT20-05
QA21-PERF-006	10k mixed worker backlog.	Run LT20-06.	Priority/convergence SLOs pass; hot tenant fairness rule passes. Trace: DOC-20 LT20-06
QA21-PERF-007	Traffic active.	Run LT20-07 Redis restart.	No authorization/site leak; caches rebuild; duplicate jobs safe. Trace: DOC-20 LT20-07
QA21-PERF-008	Traffic active across replicas.	Run LT20-08 API replica termination.	Availability within SLO; no sticky-session or local-state dependency. Trace: DOC-20 LT20-08
QA21-PERF-009	Search traffic plus normal public traffic.	Run LT20-09 search outage/recovery.	Non-search pages remain available; search degrades and rebuilds same-site. Trace: DOC-20 LT20-09
QA21-PERF-010	Concurrent writes and idempotent commands.	Run LT20-10 DB failover/connection reset.	Ambiguous writes do not double-commit; version/idempotency protect replay. Trace: DOC-20 LT20-10
QA21-PERF-011	Media transformations backlogged.	Run LT20-11.	Publish/cache queues unaffected; private original never becomes public fallback. Trace: DOC-20 LT20-11; DOC-17
QA21-PERF-012	Telemetry enabled during all load tests.	Validate pool/CPU/memory/Redis/worker/latency metrics and alert conditions.	Numeric capacity/alert/retention requirements pass; tests never weaken rate limits/RLS/security. Trace: DOC-20 QA20-003..071

23. Backup, PITR, Disaster-Recovery and Recovery Suite

Recovery evidence is produced in an isolated restore environment using real backup artifacts; tabletop discussion alone is insufficient.

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
QA21-DR-001	Production-equivalent encrypted PostgreSQL backup + WAL archive.	Run LT20-12 quarterly PITR to random point within last 7d.	Measured RPO <=5m and RTO <=60m; 35-day recoverable window policy intact. Trace: DOC-20 QA20-048..050; LT20-12
QA21-DR-002	Restored DB.	Run post-restore integrity gate.	RLS/FORCE RLS, tenant separation, 69 permission seeds, Post states/versions and audit integrity pass before readiness. Trace: DOC-20 QA20-050
QA21-DR-003	Private media backup fixture across multiple sites/same filenames.	Restore media originals.	RPO <=15m / RTO <=2h; correct owner objects; recovered originals remain private. Trace: DOC-20 QA20-051,052,058

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
QA21-DR-004	Empty Redis/search/sitemap/derived variant stores.	Rebuild solely from authoritative sources.	Completes within DOC-20 RTO matrix; no stale snapshot written back as canonical truth. Trace: DOC-20 QA20-053,059
QA21-DR-005	Catastrophic session-store loss.	Recover platform and attempt old session artifacts.	No stale session resurrected; affected users reauthenticate. Trace: DOC-20 QA20-054; DOC-19
QA21-DR-006	Broker durability/recovery scenario.	Restart/fail broker within defined disaster scenario.	Durable accepted work meets <=5m RPO / <=30m RTO and remains idempotent. Trace: DOC-20 QA20-055
QA21-DR-007	Full-platform game day.	Restore canonical/security ownership first, then derived systems; exercise all 14 sites + Admin.	Every site isolated/available within applicable RTO; no revoked secret/session revived; background convergence succeeds. Trace: DOC-20 QA20-056,057,060
QA21-DR-008	Deleted media/backups.	Inspect 7-day primary purge eligibility and 35-day inaccessible backup residual expiration.	Current usage/status/hold gates respected; historical blobs never become public/active. Trace: DOC-20 QA20-061,070; DOC-17

24. Regression and Frozen-Count Guard Suite

These tests detect silent architecture drift even when individual feature tests still pass.

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
QA21-REG-001	Candidate repository/build artifacts.	Assert frozen estate counts.	14 sites, 19 Admin modules, 36 Admin pages, 14 public Page IDs, 161 capabilities, 69 permission keys remain exact. Trace: DOC-01..06
QA21-REG-002	Database schema/seed.	Assert 33 canonical tables, 20 site-owned FORCE-RLS tables and 69 permission rows.	Counts and scope split exact; no hidden QA/product table. Trace: DOC-08
QA21-REG-003	ArticleBody schemas/renderers.	Enumerate block types end-to-end Admin->DB->PageData->frontend.	Exactly 12 approved block types; no raw/custom HTML/script/style escape hatch. Trace: DOC-07; DOC-12/13; DOC-15
QA21-REG-004	Frontend dependency graph.	Scan imports/packages for cross-site app dependencies.	No site-to-site application import; only approved shared contracts/client/utilities. Trace: DOC-14 QA14
QA21-REG-005	DOC-15 RC7 PDF/YAML.	Run full contract sync check.	112 ops/85 paths/217 schemas/1,297 refs; zero drift/unresolved refs. Trace: DOC-15 RC7
QA21-REG-006	DOC-16 implementation fixtures.	Run golden SEO/search/sitemap/redirect vectors before/after candidate changes.	No ranking/fallback/precedence/indexability drift. Trace: DOC-16 104 assertions
QA21-REG-007	DOC-17 processing registry.	Enumerate variant keys and public baseline behavior.	Fixed 16-key derivative catalog; private original never public; native_webp baseline unchanged. Trace: DOC-17
QA21-REG-008	DOC-18 registries.	Enumerate queues/jobs/retry profiles/Redis prefixes and dedupe states.	Exactly 6 logical queues, 11 canonical job types, 7 retry profiles, frozen namespace/state semantics. Trace: DOC-18
QA21-REG-009	DOC-19 auth/security contract.	Enumerate API15-AUTH operations and security constants.	11 auth operations align with DOC-19 mechanics; no bypass/test-only credential path. Trace: DOC-15 RC7; DOC-19

QA21 ID	Preconditions / matrix	Procedure	Expected result / release failure + trace
QA21-REG-010	Candidate release diff.	Compare changed behavior/schema/permissions/pages/routes against authorities.	No unapproved new feature, permission, capability, DB object, endpoint, page, tenant context, media variant or search rule. Trace: DOC-01..20 change-control invariants

25. Release Execution Pipeline and Stop-the-Line Rules

Gate	Required suites	Exit condition
G21-0 Contract/static	REG + API schema/static import checks	No frozen-count/OpenAPI/import drift.
G21-1 Isolation/data	ISO + DB	100% PASS; any leak stops all later gates.
G21-2 Authorization/security	PERM + SEC	100% PASS; no bypass/secret/privacy failure.
G21-3 Functional/runtime	ADM + PUB + CONC + LIFE + MED + SEO + WRK	100% PASS across required parameter matrices.
G21-4 Accessibility	A11Y	All frozen WCAG/semantic/responsive assertions PASS.
G21-5 Performance	PERF	All DOC-20 numeric SLO/fairness/load gates PASS.
G21-6 Recovery	DR	Required PITR/media/derived/full-DR drill evidence current and PASS.
G21-7 Final regression	REG smoke + failure-neighbor reruns	No unresolved RB-BLOCKER/RB-FAIL/BLOCKED/FLAKY release test.

STOP-THE-LINE

If a test proves tenant crossover, authorization bypass, unpublished/Preview leakage, secret exposure, lost update, duplicate/early publish, broken PITR isolation, or corruption of canonical data, execution is stopped and the candidate is rejected immediately. Later successful suites cannot offset the failure.

26. Cross-Document Consistency Audit

Authority	Result	DOC-21 outcome
DOC-01..03	PASS	QA21 verifies existing estate/scope only; no new capability/feature.
DOC-04	PASS	182 directed pair isolation methodology strengthens verification without changing SiteContext semantics.
DOC-05/06	PASS	14 public + 36 Admin page counts preserved.
DOC-07/08	PASS	Lifecycle/concurrency/RLS tests use existing states/fields/tables only.
DOC-09	PASS	Routing tests consume precedence/canonical/404/410 exactly.
DOC-10/11	PASS	Admin/public QA seeds become executable matrices; states not redefined.
DOC-12/13	PASS	WCAG 2.2 AA and PageData semantics are tested as frozen.
DOC-14	PASS	Independent frontend/import-boundary tests preserve 14-app model.
DOC-15 RC7	PASS	OpenAPI executable counts and 11 AUTH operations tested exactly.
DOC-16 RC4	PASS	104 SEO/search/sitemap/redirect assertions

Authority	Result	DOC-21 outcome
		automated without algorithm change.
DOC-17 RC2	PASS	Media privacy/baseline/variant behavior tested unchanged.
DOC-18 RC2	PASS	RC2 fingerprint/dedupe recovery semantics explicitly regression-tested.
DOC-19 RC2	PASS	Auth/session/security mechanics and resolved RC7 transport alignment tested.
DOC-20 RC1	PASS	LT20-01..12 and numeric SLO/RPO/RTO become executable release gates.

AUDIT RESULT

No genuine contradiction or blocking missing dependency was found in DOC-01 through DOC-20 for DOC-21 scope. Exact site-specific visual-design conformance remains later PDF-26/27 authority; DOC-21 here tests the already-frozen responsive/accessibility semantics only, so that later visual work does not create a current dependency gap.

27. Traceability and Coverage Ledger

Trace ID	Upstream	Locked QA21 coverage
TR21-001	DOC-01..03	REG-001/010 + all functional suites preserve exact estate/capability scope.
TR21-002	DOC-04 ISO-REQ-001..075	ISO-001..016 + DB-001..004 + WRK-002 + SEC tenant tests.
TR21-003	DOC-05 ACC-001..040 / DOC-06 AC-001..050	PUB-001..010 + ADM-005 + REG-001.
TR21-004	DOC-07 ACC-CNT-001..052	CONC-001..012 + LIFE-001..010 + PUB-002/006.
TR21-005	DOC-08 DB-ACC-001..088	DB-001..010 + ISO relationship tests + DR-002.
TR21-006	DOC-09 ACC-001..073	PUB-001/005/006 + SEO-004/012.
TR21-007	DOC-10 AC10-001..076 + page QA families	ADM-001..010 + PERM + CONC/LIFE/MED/SEO operational cases.
TR21-008	DOC-11 AC11-001..056 + QA11 families	PUB-001..010 + A11Y-010.
TR21-009	DOC-12 QA12-001..074	A11Y-001..010 + PUB rendering cases.
TR21-010	DOC-13 AC-001..072	API-004/007 + PUB + SEO + MED public projections.
TR21-011	DOC-14 QA14-001..036	PUB-010 + REG-004 + A11Y-010.
TR21-012	DOC-15 RC7 AC15-001..070	API-001..010 + PERM + SEC + REG-005/009.
TR21-013	DOC-16 RC4 104 QA16 assertions	SEO-001..012 + PUB-008 + WRK-011 + REG-006.
TR21-014	DOC-17 RC2 QA17-001..083	MED-001..010 + ISO-005 + PERF-011 + DR-003/008.
TR21-015	DOC-18 RC2 QA18-001..069	WRK-001..012 + LIFE-003..007 + PERF-005..010.
TR21-016	DOC-19 RC2 QA19-001..067	SEC-001..016 + PERM + LIFE-008 + DR-005/007.
TR21-017	DOC-20 RC1 QA20-001..072 + LT20-01..12	PERF-001..012 + DR-001..008 + DB-010.

27.1 DOC-16 normative family coverage

DOC-16 family	Upstream range	QA21 execution
SEO	QA16-SEO-01..20	QA21-SEO-001..007 plus PUB/REG checks
Sitemap	QA16-SM-01..14	QA21-SEO-008..009 + WRK sitemap convergence
Search	QA16-SRCH-01..20	QA21-SEO-010..011 + ISO-008 + CONC-011
Redirect	QA16-RED-01..16	QA21-SEO-012 + ISO-007 + PUB-006

DOC-16 family	Upstream range	QA21 execution
Data	QA16-DATA-01..12	QA21-API-004 + SEO public projection tests
API	QA16-API-01..12	QA21-API suite + PERM SEO field checks
Isolation	QA16-ISO-01..10	QA21-ISO-007..009 + SEO cross-site cases

28. Freeze-Readiness Checklist

- FZ21-01 Authority boundary accepted: DOC-21 tests but does not redefine DOC-01..20.
- FZ21-02 Deterministic QA21 family/numbering grammar accepted.
- FZ21-03 ENV21-UNIT/INT/E2E/LOAD/DR evidence roles accepted.
- FZ21-04 All 14-site directed isolation matrices and collision fixtures accepted.
- FZ21-05 RLS/DB integration gate accepted.
- FZ21-06 DOC-15 RC7 executable OpenAPI contract gate accepted.
- FZ21-07 Permission attack/direct-invocation gate accepted.
- FZ21-08 Admin multi-tab/site-switch/stale-response gate accepted.
- FZ21-09 Public 14-page/14-app runtime gate accepted.
- FZ21-10 Concurrency/autosave/idempotency race gate accepted.
- FZ21-11 Publishing/scheduling/Preview lifecycle gate accepted.
- FZ21-12 Media privacy/variant/delete/replace gate accepted.
- FZ21-13 SEO/search/sitemap/redirect deterministic gate accepted.
- FZ21-14 Worker/dedupe/Redis/reconciliation gate accepted.
- FZ21-15 Security/auth/session/MFA/CSRF/CORS/audit/privacy gate accepted.
- FZ21-16 WCAG 2.2 AA/responsive semantic gate accepted.
- FZ21-17 DOC-20 LT20-01..12 performance/fairness gate accepted.
- FZ21-18 PITR/media/derived/full-DR recovery gate accepted.
- FZ21-19 Upstream coverage ledger accepted as complete for release-blocking baselines.
- FZ21-20 Cross-document audit has zero unresolved blockers; all required QA21 tests must PASS with evidence before release.

Appendix A. Canonical QA21 Suite Registry

Family	Canonical IDs in this RC	Primary environment	Release class
ISO	QA21-ISO-001..016	INT/E2E	Release-blocking
DB	QA21-DB-001..010	INT/DR	Release-blocking
API	QA21-API-001..010	INT	Release-blocking
PERM	QA21-PERM-001..012	INT/E2E	Release-blocking
ADM	QA21-ADM-001..010	E2E	Release-blocking
PUB	QA21-PUB-001..010	E2E	Release-blocking
CONC	QA21-CONC-001..012	INT/E2E	Release-blocking
LIFE	QA21-LIFE-001..010	INT/E2E	Release-blocking
MED	QA21-MED-001..010	INT/E2E	Release-blocking
SEO	QA21-SEO-001..012	INT/E2E	Release-blocking
WRK	QA21-WRK-001..012	INT/LOAD	Release-blocking
SEC	QA21-SEC-001..016	INT/E2E	Release-blocking
A11Y	QA21-A11Y-001..010	E2E	Release-blocking
PERF	QA21-PERF-001..012	LOAD	Release-blocking

Family	Canonical IDs in this RC	Primary environment	Release class
DR	QA21-DR-001..008	DR	Release-blocking
REG	QA21-REG-001..010	UNIT/INT/E2E	Release-blocking

REGISTRY TOTAL

This RC defines 180 deterministic QA21 IDs. Parameterized child executions (including 182 directed site-pair instances where applicable) do not consume new QA IDs; they are execution instances of the canonical invariant.

Appendix B. Upstream Release-Acceptance Coverage Ledger

Upstream release baseline	Coverage status	Primary QA21 families
DOC-04 ISO-REQ-001..075	COVERED	ISO / DB / WRK / SEC
DOC-05 ACC-001..040	COVERED	PUB / ISO / REG
DOC-06 AC-001..050	COVERED	ADM / PERM / REG
DOC-07 ACC-CNT-001..052	COVERED	CONC / LIFE / PUB
DOC-08 DB-ACC-001..088	COVERED	DB / ISO / DR
DOC-09 ACC-001..073	COVERED	PUB / SEO / ISO
DOC-10 AC10-001..076 + page QA families	COVERED	ADM / PERM / CONC / LIFE / MED / SEO
DOC-11 AC11-001..056 + page QA families	COVERED	PUB / A11Y / ISO
DOC-12 QA12-001..074	COVERED	A11Y / PUB
DOC-13 AC-001..072	COVERED	API / PUB / SEO / MED
DOC-14 QA14-001..036	COVERED	PUB / A11Y / REG
DOC-15 RC7 AC15-001..070	COVERED	API / PERM / SEC / REG
DOC-16 RC4 104 QA16 assertions	COVERED	SEO / API / ISO / WRK
DOC-17 RC2 QA17-001..083	COVERED	MED / ISO / PERF / DR
DOC-18 RC2 QA18-001..069	COVERED	WRK / LIFE / PERF / ISO
DOC-19 RC2 QA19-001..067	COVERED	SEC / PERM / LIFE / DR
DOC-20 RC1 QA20-001..072 + LT20-01..12	COVERED	PERF / DR / DB

DOC-01 through DOC-03 contain foundational scope/capability constraints rather than one uniform acceptance-ID sequence. Their normative invariants are covered by REG-001/010 and by every suite's no-scope-creep checks. This ledger intentionally does not invent replacement IDs for upstream requirements.

Appendix C. Evidence Artifact Manifest

Evidence ID	Artifact	Required when
EV21-HTTP	Sanitized request/response transcript with operation ID, status, request ID	API/auth/isolation/functional tests
EV21-DB	Scoped SQL assertion/report showing role, transaction SiteContext and result	RLS/data/concurrency/restore tests
EV21-UI	Screenshot + URL/Host + DOM/accessibility state where visual runtime matters	Admin/public/a11y tests
EV21-A11Y	Automated scan output + keyboard/focus checklist	A11Y suite
EV21-QUEUE	Broker delivery/attempt/DLQ/replay trace keyed by job ID	WRK/LIFE tests
EV21-REDIS	Sanitized key/state/TTL observation showing tenant namespace	WRK/cache tests
EV21-SEARCH	Index generation/query result snapshot with site	SEO/search tests

Evidence ID	Artifact	Required when
	identity	
EV21-SITEMAP	Fetched sitemap index/shards + generated manifest	SEO/sitemap tests
EV21-MEDIA	Object/variant manifest, MIME/dimensions/metadata-sanitization evidence	MED tests
EV21-AUDIT	Authorized audit-event correlation proof	Security/destructive/workflow tests
EV21-METRIC	Time-bounded metrics dashboards/export + percentile calculation	PERF tests
EV21-TRACE	Distributed trace/correlation record with secrets removed	Fault/concurrency/performance investigations
EV21-BACKUP	Backup identifier/checksum/verification log	DR tests
EV21-RESTORE	Restore timeline, RPO/RTO calculation and post-restore integrity report	DR/PITR tests
EV21-DEFECT	Defect record with root cause/fix/build/rerun evidence	Any FAIL/FLAKY

EVIDENCE PRIVACY

Evidence must obey DOC-19 privacy/secret rules. Do not attach passwords, MFA seeds, raw reset/invitation/Preview grants, signed private URLs, object-store credentials, article-body dumps beyond the minimum fixture markers, or raw production personal data.

Appendix D. One-Page Release Execution Reference

Stage	Execute	Hard pass rule
0	Freeze candidate + fixture manifest + OpenAPI/schema static checks	Immutable candidate identified; REG/API static checks PASS.
1	ISO + DB	All directed isolation/RLS instances PASS; zero leak.
2	PERM + SEC	No authorization/session/MFA/CSRF/CORS/secret bypass.
3	ADM + PUB + CONC + LIFE + MED + SEO + WRK	All functional/race/derived-state matrices PASS.
4	A11Y	All frozen WCAG/responsive semantic tests PASS.
5	PERF	LT20-01..11 + numeric SLO/fairness thresholds PASS.
6	DR	PITR/media/derived/full-DR evidence meets RPO/RTO; LT20-12 PASS.
7	Final regression + evidence completeness	No FAIL/BLOCKED/FLAKY; every QA21 result linked to evidence and source trace.

FINAL RELEASE RULE

Release = every applicable canonical QA21 ID PASS across its required parameter instances + complete evidence + zero unresolved release-blocking defect. No later document may weaken these expected outcomes without controlled DOC-21 revision.